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## **Electricity Bill Forecaster Model**

### **Introduction**

Electricity has become one of the most essential resources in modern society, powering homes, industries, educational institutions, and businesses. With increasing energy consumption and rising electricity costs, it has become important for consumers to monitor and manage their electricity usage efficiently. Predicting future electricity bills can help individuals and organizations plan their expenses, identify abnormal consumption patterns, and adopt energy-saving measures.

Machine Learning provides powerful techniques for analyzing historical electricity consumption data and forecasting future usage. Among various predictive algorithms, Multiple Linear Regression is one of the most widely used methods due to its simplicity and effectiveness in identifying relationships between multiple independent variables and a target variable. In this project, an Electricity Bill Forecaster is developed using Multiple Linear Regression to predict monthly electricity consumption costs based on historical usage data and other influencing factors.

The proposed system aims to assist users in estimating future electricity expenses by analyzing past consumption patterns. Such predictions can help consumers make informed decisions regarding energy utilization and budget planning. The model can also be beneficial for electricity providers and energy management systems in understanding consumption trends and improving resource allocation.

### **Problem Statement**

The main objective of the "Electricity Bill Forecaster" project is to develop a predictive model that estimates the monthly electricity bill of a consumer based on historical electricity consumption data and related factors. Electricity usage depends on several variables such as the number of units consumed, seasonal changes, household occupancy, appliance usage, and weather conditions. By studying these variables, the system can learn patterns and generate accurate forecasts for future electricity costs.

The problem can be defined as follows: Given a dataset containing historical electricity consumption records and associated billing information, build a Multiple Linear Regression model that can predict the expected electricity bill for a future month. The model should analyze the relationship between independent variables and electricity cost while minimizing prediction errors.

The primary goal of this project is to provide a simple, accurate, and interpretable forecasting system that can assist consumers in monitoring their electricity expenses. The project also demonstrates the practical application of machine learning techniques in solving real-world energy management problems.

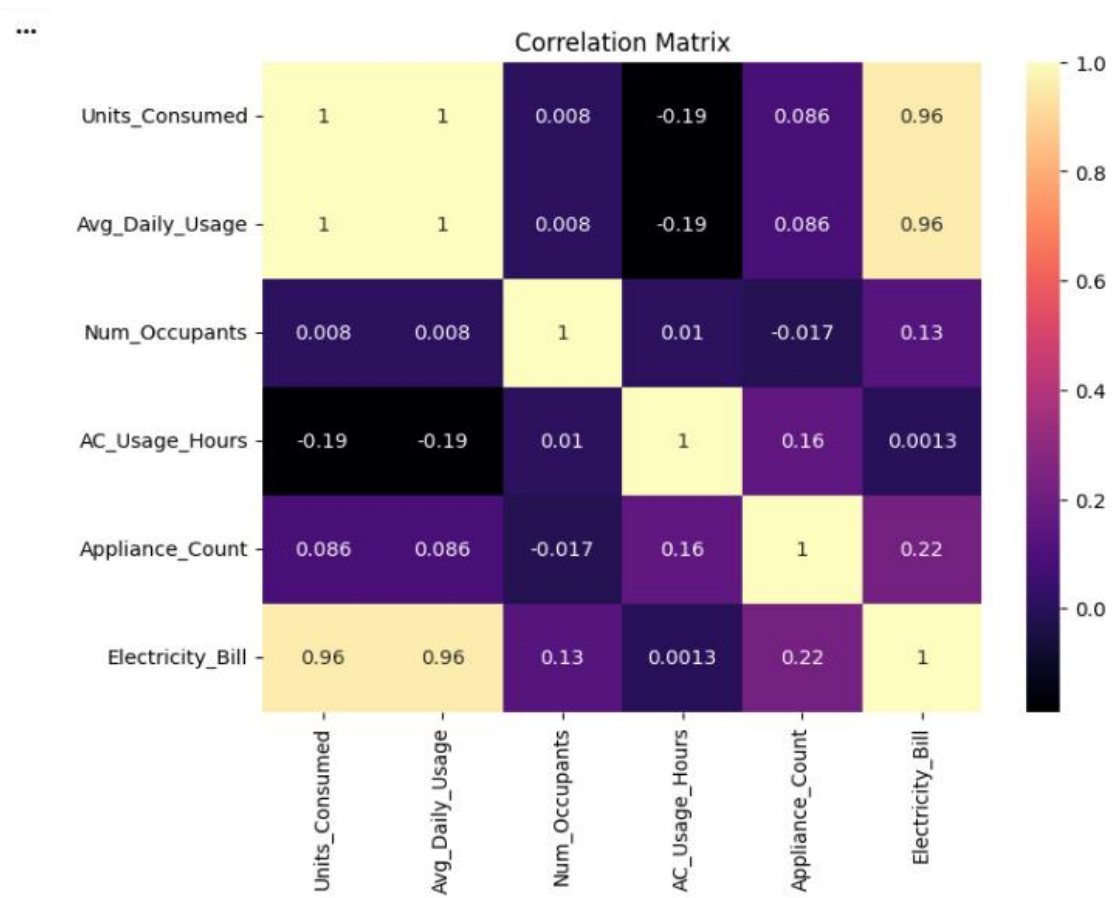
## Results and Discussion

The Multiple Linear Regression model was trained using historical electricity consumption data and tested on unseen records to evaluate its prediction capability. Various preprocessing techniques such as data cleaning, handling missing values, and feature selection were performed before model training. Exploratory Data Analysis revealed a strong positive relationship between electricity units consumed and the total monthly bill amount.

The model successfully identified important factors affecting electricity costs. It was observed that households with higher appliance usage and greater monthly energy consumption generated larger electricity bills. Seasonal variations also influenced electricity usage, particularly during summer months when cooling appliances were used more frequently.

Performance evaluation metrics such as *Mean Absolute Error (MAE)*, *Mean Squared Error (MSE)*, and *R<sup>2</sup> Score* were used to assess model accuracy. The results indicated that the model was able to predict electricity bills with a high level of reliability. Graphical comparisons between actual and predicted bill values showed that the regression model effectively captured consumption trends and produced forecasts close to real-world observations.

The findings demonstrate that Multiple Linear Regression can serve as an efficient forecasting tool for electricity bill prediction and can help consumers better understand the factors influencing their energy expenses.



```

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics

X = data[['Units_Consumed',
          'Avg_Daily_Usage',
          'Num_Occupants',
          'AC_Usage_Hours',
          'Appliance_Count']]

y = data['Electricity_Bill']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

pred = model.predict(X_test)

print("MSE:", metrics.mean_squared_error(y_test, pred))

accuracy = model.score(X_test, y_test)
print("R2 Score:", accuracy)

... MSE: 14534.346061937318
    R2 Score: 0.9713488009716075

```

## Conclusion

In this project, a machine learning-based Electricity Bill Forecaster was developed using the Multiple Linear Regression algorithm. The system was designed to predict monthly electricity costs by analyzing historical consumption data and various contributing factors. Through data preprocessing, feature analysis, and model training, the developed solution was able to generate accurate and reliable bill predictions.

The project highlights the effectiveness of regression techniques in forecasting energy consumption-related expenses. By providing advance estimates of future electricity bills, the system can assist consumers in budgeting, monitoring energy usage, and implementing energy-saving practices. The results obtained indicate that Multiple Linear Regression is a suitable approach for electricity bill forecasting due to its simplicity, interpretability, and computational efficiency.

In the future, the project can be extended by incorporating real-time smart meter data, weather information, and advanced machine learning algorithms such as Random Forest, XGBoost, and Artificial Neural Networks. Additionally, a web-based or mobile application can be developed to provide users with an interactive platform for electricity bill prediction and energy management.

**Reference:** A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, Scikit-Learn Documentation for Linear Regression Models, Research papers on electricity consumption forecasting and energy analytics, Historical electricity consumption datasets available from open data repositories.